

Certified Penetration Testing Professional

Methodology: Binary Analysis and Exploitation

Penetration Tester:			
Organization:			
Date:		**Location:**	

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Test 1: Binary Coding Concepts

Test 1.1: Analyze Binaries

Target Organization	
URL	
Application/Program Tested	
Commands Used	
Analyzed Binaries Successfully?	☐ Yes ☐ No

Information Collected

	Type of Data	Headers	Compositions	Strings	Entry Points
1.					
2.					
3.					
4.					
5.					
6.					
7.					
8.					
9.					
10.					

Data Format	
Any Other Information Collected	1. _____ 2. _____ 3. _____ 4. _____ 5. _____

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Tools/Services Used	1. 2. 3. 4. 5.

Results Analysis:

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Test 2: Binary Analysis Methodology

Test 2.1: Binary Discovery

Target Organization	
URL	
Application/Program Tested	
Commands Used	
List of Discovered Binaries	1. 2. 3. 4. 5.
Tools/Services Used	1. 2. 3. 4. 5.

Results Analysis:

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Test 2.2: Information Gathering

Target Organization			
URL			
Application/Program Tested			
Commands Used			
Information Gathered Successfully?		☐ Yes	☐ No
Is the file executable?		☐ Yes	☐ No
Is the file binary?		☐ Yes	☐ No
Architecture of the binary	☐ X86	☐ X86-64	
Is the binary stripped of its symbol table?		☐ Yes	☐ No
Can any useful strings be identified within the binary?		☐ Yes	☐ No
Is there a running process associated with this binary?		☐ Yes	☐ No
Is the hash a known malicious file hash?		☐ Yes	☐ No
Can any useful function names be identified?		☐ Yes	☐ No
Can any libraries used to be identified?		☐ Yes	☐ No
Does the application take any input?		☐ Yes	☐ No
Specify the input taken by the Application	1. 2. 3. 4. 5.		
Does the application safely manage memory?		☐ Yes	☐ No
Does the application use up-to-date libraries or third-party frameworks?		☐ Yes	☐ No

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Specify the libraries and third-party frameworks used by the Application	1. 2. 3. 4. 5.
How is the application compiled?	
Are there any noticeable strings containing sensitive data such as hardcoded passwords?	☐ Yes ☐ No
Specify sensitive data stored in strings	1. 2. 3. 4. 5.
Secure Hash Algorithm (SHA) hash of the binary	
Original Programming Language used	
Binary Written to Disk	

Information Collected					
	Type of Data	**Headers**	**Compositions**	**Strings**	**Entry Points**
1.					
2.					
3.					
4.					
5.					
6.					

Data Format	

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Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.3: Static Analysis

Target Organization			
URL			
Commands Used			
Is Static Analysis Successful?		☐ Yes	☐ No
Information Extracted	1. 2. 3. 4. 5.		
Tools/Services Used	1. 2. 3. 4. 5.		

Results Analysis:

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Test 2.4: Dynamic Analysis

Target Organization		
URL		
Commands Used		
Dynamic Analysis Successful?	☐ Yes	☐ No
Information Extracted	1. 2. 3. 4. 5.	
Tools/Services Used	1. 2. 3. 4. 5.	

Results Analysis:

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Test 2.5: Defeat the No-execute Stack

Target Organization			
URL			
Commands Used			
Is Defeating the No-execute Stack Successful?		☐ Yes	☐ No
Program			
External Location			
Address of the Environment Variable			
Tools/Services Used	1. 2. 3. 4. 5.		

Results Analysis:

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Test 2.6: Attack using ROP

Target Organization		
URL		
Commands Used		
Is Attack using ROP Successful?	☐ Yes	☐ No
Building Custom ROP Gadget Tool Successful?	☐ Yes	☐ No
Gadgets	1. 2. 3. 4. 5.	
Addresses of the Gadgets	1. 2. 3. 4. 5.	
Result of the Attack		
Tools/Services Used	1. 2. 3. 4. 5.	

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